

Unique domination in bipartite graphs of order $3\gamma + 1$

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Abstract

A graph with no isolated vertices and a unique minimum dominating set of cardinality γ has at least 3γ vertices. We determine the maximum number of edges in a bipartite graph one vertex above this threshold. For every $\gamma \geq 2$, the maximum at order $3\gamma + 1$ is $\gamma(\gamma + 7)/2$, and a connected graph attains it. For $\gamma \geq 4$, we characterize the extremal graph uniquely up to isomorphism. The construction exceeds the corresponding bound conjectured by Koch and Narayan by $\lfloor \gamma/2 \rfloor - 1$ edges; in particular, we recover a previously reported 13-vertex, 22-edge counterexample. The proof is elementary. It combines the two exterior private neighbours forced by uniqueness with replacement arguments on six and seven vertices, followed by a rigidity analysis of the equality case.

Keywords: unique minimum dominating set; bipartite graph; extremal size; private neighbour.

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1 Introduction

All graphs in this paper are finite and simple. A set $D \subseteq V(G)$ is *dominating* if every vertex outside D has a neighbour in D . The domination number $\gamma(G)$ is the minimum cardinality of a dominating set. A graph has a *unique minimum dominating set* if exactly one dominating set has cardinality $\gamma(G)$. Throughout, uniqueness refers to minimum cardinality, not to inclusion-minimality. Write $n(G) = |V(G)|$ and $e(G) = |E(G)|$.

Fischermann, Rautenbach and Volkmann [1] investigated the maximum size of graphs with a unique minimum dominating set. Fraboni and Shank [2] subsequently settled their conjectured size bound when the domination number is two. Koch and Narayan [4] considered the bipartite problem and proposed a bound depending on the order and the domination number. They proved their bound for domination number two and at order $n = 3\gamma$. Erbacher's public artifacts [3] already give the 13-vertex, 22-edge counterexample, further examples, and finite extremal computations. The result below recovers that example and proves the sharp boundary maximum and equality structure for general domination number. Koch and Narayan also identify the unique extremal graph at $(n, \gamma) = (10, 3)$ in their Figure 1 discussion.

The order restriction is intrinsic. If D is the unique minimum dominating set of a graph with no isolated vertices, every vertex of D has at least two exterior private neighbours with respect to D . These neighbours are distinct for distinct members of D , and hence $n(G) \geq 3\gamma(G)$. We recall the short argument in Lemma 3.1.

We examine the next possible order. Let $\mathcal{B}_\gamma$ be the class of bipartite graphs with no isolated vertices, of order $3\gamma + 1$, whose minimum dominating set is unique and has cardinality γ . No connectedness assumption is imposed in the definition.

Theorem 1.1. *For every integer $\gamma \geq 2$,*

$$\max_{G \in \mathcal{B}_\gamma} e(G) = \frac{\gamma(\gamma + 7)}{2}.$$

The maximum is attained by a connected graph. For $\gamma \geq 4$, equality holds if and only if

$$G \cong H(\lceil \gamma/2 \rceil, \lfloor \gamma/2 \rfloor),$$

where $H(p, q)$ is the graph defined in Section 2.

At order $3\gamma + 1$, Conjecture 1 of [4] specializes to

$$e(G) \leq 2\gamma + 2ab + 2a + 1, \quad a = \lceil \gamma/2 \rceil, \quad b = \lfloor \gamma/2 \rfloor. \quad (1)$$

The summation term in the conjecture is empty at this order. Since

$$\frac{\gamma(\gamma + 7)}{2} - (2\gamma + 2ab + 2a + 1) = b - 1, \quad (2)$$

Theorem 1.1 has the following consequence.

Corollary 1.2. *For every $\gamma \geq 4$, there is a connected bipartite graph on $3\gamma + 1$ vertices with a unique minimum dominating set of cardinality γ that violates (1). The excess is $\lfloor \gamma/2 \rfloor - 1$. In particular, at $\gamma = 4$ the extremal graph has 13 vertices and 22 edges, whereas (1) gives 21.*

The additional vertex explains the change from the threshold case $n = 3\gamma$. It may be adjacent to several members of the unique minimum dominating set. Our extremal graph uses this possibility. The proof of the upper bound keeps track of whether the additional vertex has one or several such neighbours; the equality argument then forces the whole construction.

2 The extremal construction

Definition 2.1. For integers $p, q \geq 1$, let $H(p, q)$ have pairwise distinct vertices

$$\begin{aligned} X &= \{x_i : 1 \leq i \leq p\}, & Y &= \{y_j : 1 \leq j \leq q\}, \\ U_i &= \{u_i^0, u_i^1\} \quad (1 \leq i \leq p), & V_j &= \{v_j^0, v_j^1\} \quad (1 \leq j \leq q), \end{aligned}$$

and one further vertex z . Its edges are exactly

$$\begin{aligned} x_i u_i^\varepsilon & \quad (1 \leq i \leq p, \varepsilon \in \{0, 1\}), \\ y_j v_j^\varepsilon & \quad (1 \leq j \leq q, \varepsilon \in \{0, 1\}), \\ u_i^\varepsilon v_j^0 & \quad (1 \leq i \leq p, 1 \leq j \leq q, \varepsilon \in \{0, 1\}), \\ z u_i^\varepsilon & \quad (1 \leq i \leq p, \varepsilon \in \{0, 1\}), \\ z y_j & \quad (1 \leq j \leq q). \end{aligned}$$

The displayed graph is bipartite, with parts

$$X \cup \bigcup_{j=1}^q V_j \cup \{z\} \quad \text{and} \quad Y \cup \bigcup_{i=1}^p U_i. \quad (3)$$

It is connected: z is adjacent to all of $Y \cup \bigcup_i U_i$, each x_i has a neighbour in U_i , and each vertex of V_j is adjacent to y_j .

Proposition 2.2. *The graph $H(p, q)$ has order $3(p + q) + 1$ and size $2pq + 4p + 3q$. Its unique minimum dominating set is $X \cup Y$, and its domination number is $p + q$.*

Proof. The order follows by counting vertices, and the five edge classes in Definition 2.1 have respective sizes $2p, 2q, 2pq, 2p, q$. Clearly $D = X \cup Y$ dominates the graph.

Every dominating set K must meet $N[x_i] = \{x_i, u_i^0, u_i^1\}$ for each i . It must also meet $N[v_j^1] = \{y_j, v_j^1\}$ for each j , since v_j^1 is a leaf. These $p+q$ sets are pairwise disjoint, so $|K| \geq p+q$. This proves $\gamma(H(p, q)) = p+q$.

Suppose now that $|K| = p+q$. It chooses exactly one vertex from each of these sets and no other vertex. In particular, z and all v_j^0 are absent. If K contains u_i^0 rather than x_i , then u_i^1 is undominated: its neighbours are x_i, z and the vertices v_j^0 . The same argument with the superscripts reversed excludes u_i^1 . Thus every x_i belongs to K and no vertex of any U_i belongs to K . Finally, choosing v_j^1 rather than y_j leaves v_j^0 undominated, because its neighbours are y_j and the vertices in the sets U_i . Hence $K = X \cup Y$. $\square$

For $a = \lceil \gamma/2 \rceil$ and $b = \lfloor \gamma/2 \rfloor$, a direct calculation gives

$$e(H(a, b)) = 2ab + 4a + 3b = \frac{\gamma(\gamma + 7)}{2}. \quad (4)$$

Proposition 2.2 therefore supplies the lower bound in Theorem 1.1, including connectedness.

3 Private neighbours and local replacements

For a dominating set D and $d \in D$, define

$$\text{epn}(d, D) = \{u \in V(G) \setminus D : N(u) \cap D = \{d\}\}.$$

We use closed neighbourhoods only when square brackets are displayed.

Lemma 3.1. *If G has no isolated vertices and D is its unique minimum dominating set, then $|\text{epn}(d, D)| \geq 2$ for every $d \in D$.*

Proof. If $\text{epn}(d, D) = \{u\}$, the set $(D \setminus \{d\}) \cup \{u\}$ dominates G : u dominates itself and d , while every other vertex outside D retains a neighbour in $D \setminus \{d\}$. This contradicts uniqueness.

If $\text{epn}(d, D) = \emptyset$ and d has a neighbour in D , then $D \setminus \{d\}$ is a smaller dominating set. Otherwise d has no neighbour in D . The no-isolate assumption gives a neighbour u outside D , and $(D \setminus \{d\}) \cup \{u\}$ is again a different dominating set of the same cardinality. All cases are impossible. $\square$

The next two lemmas concern induced subgraphs that arise after two members of D are removed. They include every possible edge between the two centers. The six-vertex argument is the local estimate used in the proof of Theorem 12 of [4]; we record its equality patterns as well.

Lemma 3.2. *Let F be a bipartite graph with parts $\{x\} \cup V$ and $\{y\} \cup U$, where $|U| = |V| = 2$. Assume that x is adjacent to both vertices of U and y to both vertices of V . If $\{x, y\}$ is the unique dominating set of cardinality at most two, then at most two edges occur among xy and the U – V edges. If exactly two occur, either xy is absent and the U – V edges form a two-edge star, or xy is present and there is exactly one U – V edge.*

Proof. Two disjoint U – V edges give an alternative dominating pair: select opposite endpoints, one from each edge. If xy is absent, three U – V edges contain two disjoint edges. Thus there are at most two, and a two-edge configuration must be a star.

Suppose xy is present. Two U – V edges either are disjoint or form a star. A star centered at $u \in U$ gives the dominating pair $\{x, u\}$; a star centered at $v \in V$ gives $\{y, v\}$. Consequently at most one U – V edge can accompany xy . $\square$

Lemma 3.3. *Let F have the same form as in Lemma 3.2, except that $|U| = 2$ and $|V| = 3$. If $\{x, y\}$ is the unique dominating set of cardinality at most two, then at most four edges occur among xy and the U – V edges.*

Proof. If xy is absent and at least five U – V edges occur, one vertex $u \in U$ is adjacent to all three vertices of V , and the other vertex u' is adjacent to at least two. Choose $v \in V$ adjacent to u' . The pair $\{u, v\}$ dominates F , a contradiction.

If xy is present, any vertex $v \in V$ adjacent to both vertices of U gives the alternative dominating pair $\{y, v\}$. Every vertex of V therefore has at most one neighbour in U . There are at most three U – V edges, and at most four edges when xy is included. $\square$

These are direct combinatorial proofs. The exhaustive checks summarized in Appendix A are supplementary and are not used as hypotheses in what follows.

4 The sharp upper bound

Let $G \in \mathcal{B}_\gamma$, and let D be its unique minimum dominating set. By Lemma 3.1, we may select two exterior private neighbours for each vertex of D . The selected pairs are disjoint and account for 2γ vertices. Since $n(G) = 3\gamma + 1$, exactly one further vertex z remains.

Choose the names of the bipartition classes so that $z \in A$, and write

$$D \cap A = \{x_1, \dots, x_p\}, \quad D \cap B = \{y_1, \dots, y_q\}, \quad p + q = \gamma.$$

Let $U_i \subseteq B$ be the selected private pair of x_i , and $V_j \subseteq A$ the selected private pair of y_j . Set

$$J = N(z) \cap D \subseteq \{y_1, \dots, y_q\}, \quad t = |J|.$$

Because D dominates z , we have $t \geq 1$, and in particular $q \geq 1$.

Privacy and bipartiteness give a complete list of possible edges:

1. the 2γ selected center-to-private edges;
2. $x_i y_j$ and the edges between U_i and V_j , for each pair (i, j) ;
3. the edges from z to J and to $\bigcup_i U_i$.

In particular, edges inside D have not been discarded: they are precisely the possible edges $x_i y_j$ in the second class.

4.1 The additional vertex has several dominators

Suppose $t \geq 2$. For each i, j , consider the induced six-vertex graph

$$G[K_{ij}], \quad K_{ij} = \{x_i, y_j\} \cup U_i \cup V_j.$$

The set $D \setminus \{x_i, y_j\}$ dominates every vertex outside K_{ij} . Each selected private vertex outside the cell retains its own center; the vertex z retains a neighbour in J because only one y -center has been removed.

If $G[K_{ij}]$ had a dominating set S of cardinality at most two other than $\{x_i, y_j\}$, then

$$(D \setminus \{x_i, y_j\}) \cup S$$

would be a dominating set of G of cardinality at most γ , distinct from D when its cardinality is γ . This contradicts minimum cardinality or uniqueness. Thus Lemma 3.2 applies to every cell. The optional edge sets of these cells are disjoint, and $\deg(z) \leq 2p + q$. Consequently

$$e(G) \leq 2\gamma + 2pq + 2p + q. \tag{5}$$

When $p = 0$, the same bound follows directly from the edge list, with an empty sum over cells.

4.2 The additional vertex has one dominator

Suppose $J = \{y_{j_0}\}$. For every $j \neq j_0$, the preceding six-vertex replacement still applies, since y_{j_0} is retained and dominates z . For each i consider instead

$$L_i = \{x_i, y_{j_0}\} \cup U_i \cup V_{j_0} \cup \{z\}.$$

In $G[L_i]$, the center y_{j_0} has the three private vertices $V_{j_0} \cup \{z\}$. Every vertex outside L_i is dominated by $D \setminus \{x_i, y_{j_0}\}$, so the same replacement argument proves that Lemma 3.3 applies.

The optional edges of $G[L_i]$ are $x_i y_{j_0}$ and the edges from U_i to $V_{j_0} \cup \{z\}$. These edge sets are disjoint for different i . The edge $z y_{j_0}$ is counted once separately. Therefore

$$\begin{aligned} e(G) &\leq 2\gamma + 1 + 2p(q-1) + 4p \\ &= 2\gamma + 1 + 2pq + 2p \\ &\leq 2\gamma + 2pq + 2p + q. \end{aligned} \tag{6}$$

If $p = 0$, only the 2γ fixed private edges and $z y_{j_0}$ occur, so the displayed inequalities remain valid.

4.3 Optimizing the bipartition

Put $d = p - q$. Since $p + q = \gamma$, both cases yield

$$e(G) \leq 2\gamma + 2pq + 2p + q = \frac{\gamma(\gamma+7)}{2} - \frac{d(d-1)}{2} \leq \frac{\gamma(\gamma+7)}{2}. \tag{7}$$

The final inequality holds because d is an integer. Equality in the numerical optimization requires $d \in \{0, 1\}$. The parity of γ then forces

$$p = \lceil \gamma/2 \rceil, \quad q = \lfloor \gamma/2 \rfloor. \tag{8}$$

Together with (4), this proves the extremal-value assertion of Theorem 1.1 for every $\gamma \geq 2$.

5 Rigidity at equality

Suppose now that $\gamma \geq 4$ and $e(G) = \gamma(\gamma+7)/2$. Equation (8) gives $q = \lfloor \gamma/2 \rfloor \geq 2$. The final inequality in (6) loses $q-1$ edges, so the case $t = 1$ cannot give equality. Every inequality in (5) must therefore be an equality. It follows that z is adjacent to every y_j and to every vertex of $\bigcup_i U_i$, and each six-vertex cell has exactly two optional edges.

We first show that D is independent. If $x_i y_j$ were an edge, then

$$(D \setminus \{x_i\}) \cup \{z\}$$

would be a different dominating set of cardinality γ . Indeed, x_i is dominated by the retained y_j , the pair U_i is dominated by z , and every other private pair retains its center. Thus no edge $x_i y_j$ occurs.

Lemma 3.2 now shows that the two edges between U_i and V_j form a star. This star cannot be centered at $u \in U_i$, because

$$(D \setminus \{x_i, y_j\}) \cup \{u, z\}$$

would dominate G . The vertex u dominates x_i and both members of V_j , while z dominates y_j and both members of U_i . All other private pairs retain their centers. Therefore both members of U_i are adjacent to exactly one vertex of V_j ; denote it by $v_j^{(i)}$.

Fix j . Suppose that the selected vertex $v_j^{(i)}$ is not constant as i varies. Choose one vertex $u_i \in U_i$ for every i , and define

$$K = \{z\} \cup \{u_i : 1 \leq i \leq p\} \cup ((D \cap B) \setminus \{y_j\}).$$

Then $|K| = 1 + p + (q - 1) = \gamma$. Every x_i is dominated by u_i ; every vertex in $\bigcup_i U_i$ and every y -center is dominated by z . The two vertices of V_j are dominated because both possible choices of $v_j^{(i)}$ occur. Every other pair V_ℓ retains y_ℓ . Hence K is a second minimum dominating set, a contradiction.

For each j , rename the members of V_j so that their common selected vertex is v_j^0 . The complete edge list is now exactly that of $H(p, q)$. Equation (8) proves the equality assertion in Theorem 1.1.

6 Consequences and further questions

The extremal values at $\gamma = 2, 3$ are 9 and 15; the graphs $H(1, 1)$ and $H(2, 1)$ attain them. The rigidity assertion proved above is restricted to $\gamma \geq 4$. Neither the theorem nor Corollary 1.2 asserts that the 13-vertex example is the smallest counterexample over all orders in the conjecture of [4].

For $\gamma \geq 4$, the additional vertex of the extremal graph has $q = \lfloor \gamma/2 \rfloor \geq 2$ neighbours in the unique minimum dominating set. Consequently this set is not an efficient dominating set. The shared vertex is forced at equality and is responsible for the additional edges over (1). At $n = 3\gamma$, there is no vertex available for this role.

A natural next problem is to determine the extremal size at $n = 3\gamma + r$ for a fixed $r \geq 2$, together with the equality cases. Selecting two private neighbours for each dominator then leaves r vertices. Replacing two centers may leave some of those vertices undominated, so their incidences with the dominating set must be retained in any local reduction. The exact threshold at which new patterns arise is not determined here.

A Supplementary exact checks

The proofs in the main text do not depend on a computer enumeration. Two separate implementations were nevertheless used to check the local lemmas, the explicit counterexample, and the first values of the construction. All tests use finite sets or integer bit masks.

For the six-vertex cell, the fixed edges are xu_0, xu_1, yv_0, yv_1 . The optional edges, in bit order, are

$$xy, u_0v_0, u_0v_1, u_1v_0, u_1v_1.$$

For the seven-vertex cell, the fixed edges are $xu_0, xu_1, yv_0, yv_1, yv_2$, and the optional edges are

$$xy, u_0v_0, u_0v_1, u_0v_2, u_1v_0, u_1v_1, u_1v_2.$$

Each implementation considers every optional-edge subset and tests every vertex subset of cardinality at most two for domination. Table 1 records the results. “Admissible” means that $\{x, y\}$ is the only dominating set of cardinality at most two. Decimal masks use the first listed edge as the least significant bit.

Cell	Patterns	Admissible	Maximum	Maximizing masks
Six vertices	32	14	2	3, 5, 6, 9, 10, 17, 20, 24
Seven vertices	128	58	4	54, 90, 108

Table 1: Exhaustive local checks. The maximum counts optional edges.

A separate verifier reads the explicit 13-vertex edge list and checks all $2^{13} = 8192$ vertex subsets. It finds no dominating set of size at most three and exactly one of size four, namely $X \cup Y$. It also checks connectedness, bipartiteness, simplicity and the edge count 22. The counts of dominating sets, by cardinality $0, 1, \dots, 13$, are

$$(0, 0, 0, 0, 1, 127, 492, 855, 876, 586, 262, 76, 13, 1).$$

The first balanced construction values are recorded in Table 2; all subsets of cardinality at most γ were checked.

γ	(p, q)	n	$e(H(p, q))$	Minimum dominating sets
2	(1, 1)	7	9	1
3	(2, 1)	10	15	1
4	(2, 2)	13	22	1
5	(3, 2)	16	30	1
6	(3, 3)	19	39	1

Table 2: Small instances of the connected extremal family.

The accompanying files `check_local_cells.py` and `local_cells_certificate.json` give all local patterns and their dominating pairs. The explicit example is in `candidate13.json`; `check_counterexample_independent.py` verifies it from that edge list. A second implementation, `audit/independent_check.py`, independently repeats the local checks and tests all 4096 saturated assemblies at $p = q = 2$. Exactly four labelled assemblies retain a unique minimum dominating set; they correspond to the independent choice of which member of each V_j is named v_j^0 , and are isomorphic to $H(2, 2)$.

References

- [1] M. Fischermann, D. Rautenbach and L. Volkmann, *Maximum graphs with a unique minimum dominating set*, Discrete Mathematics **260** (2003), 197–203.
- [2] M. Fraboni and N. Shank, *Maximum graphs with unique minimum dominating set of size two*, Australasian Journal of Combinatorics **46** (2010), 91–99. https://ajc.maths.uq.edu.au/pdf/46/ajc_v46_p091.pdf.
- [3] J. Erlbacher, *Demonstrandum—wave-1 verified artifacts: Koch–Narayan counterexample package*, public research artifacts, 2026. Pinned write-up, commit 94db9ed50d48 (13 June 2026).
- [4] G. Koch and D. Narayan, *Maximal bipartite graphs with a unique minimum dominating set*, arXiv:2511.01719v1, 3 November 2025. <https://arxiv.org/abs/2511.01719v1>.